

Design FMEA - Qualitative (VBF-T5R1SINGLEMODEL-DFMEA-001)

> Case: t5_r1_singlemodel. Qualitative version, based on simulation signals; severity/occurrence are three-level ratings (high/medium/low) anchored to the magnitude of the simulated value vs threshold. RPN = simplified SxO qualitative matrix. Complete FMEA (process/supplier failures) is N/A beyond the pure-simulation boundary.

Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
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Negative electrode lithium plating (4C fast charge)	anode potential < 0 V during high-rate charge	`anode_potential_v` min = +0.0215 V at 4C/45 °C — margin +21.5 mV (n			
Thermal runaway risk (temperature over limit)	insufficient cooling / high-rate heat generation	T_max 330.50 K vs 333.15 K — margin 2.65 K (thinnest margin of the three cr			
Electrolyte oxidative decomposition at high voltage	electrolyte HOMO vs cathode upper cut-off 4.2 V	**not verified in this case** — funnel_voting OFF ablation removed the			
Insufficient capacity / energy density	mass overshoot or porosity-slack assumption break	ED 533.18 Wh/kg vs 500.94 — margin +32.2; capacity is negative-limited (N/P ≈ 1			
Cell mass under-count	electrolyte excluded from contract mass caliber	contract mass 34.46 g vs 41.08 g with electrolyte (literature density) low — disclosed in spec/dat			
Anode oversize absent (N/P ≈ 1.00)	mass-first design removed conventional N/P > 1 headroom	negative electrode binds capacity (R2: −8% neg thickness → −7.9% capacit			

Conclusion

Highest-risk items: (1) electrolyte oxidative stability unverified (no HOMO data under the funnel_voting OFF ablation; true-compute skipped) — recommend xTB HOMO screening + true-compute endorsement as follow-up; (2) N/P ≈ 1.00 negative-limited design — plating robustness relies on the kinetic bridge, validated only at the simulated fresh-cell condition. All mitigations listed above are implemented in the final design (archN_robust) or disclosed as limitations. Complete FMEA incl. process/supplier failures: N/A (beyond pure-simulation boundary).